Lab Manual - LAB 5 & 6

Semester V – Fall

Group A and B

Course Title: Computer Networks

Bachelor of Science in Software Engineering

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LAB 05 & 06 — IPv4 Subnetting and Subnet Design

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IPv4 Subnetting and Subnet Design

Learning Objectives

By the end of this lab, students will be able to:

- Understand the purpose of subnetting.
- Understand notation and subnet masks.
- Identify network and host portions of an IPv4 address.
- Calculate the number of subnets.
- Calculate the number of hosts per subnet.
- Find the network ID.
- Find the broadcast address.
- Determine the usable host range.
- Design subnet ranges according to host requirements.

Introduction to Subnetting

In previous labs, we worked with IPv4 networks such as: 192.168.10.0/24. A /24 network has a Subnet Mask: 255.255.255.0. It contains:

- 256 total addresses
- 254 usable host addresses

For a small network, this may be sufficient. However, suppose a university has one network:

192.168.10.0/24

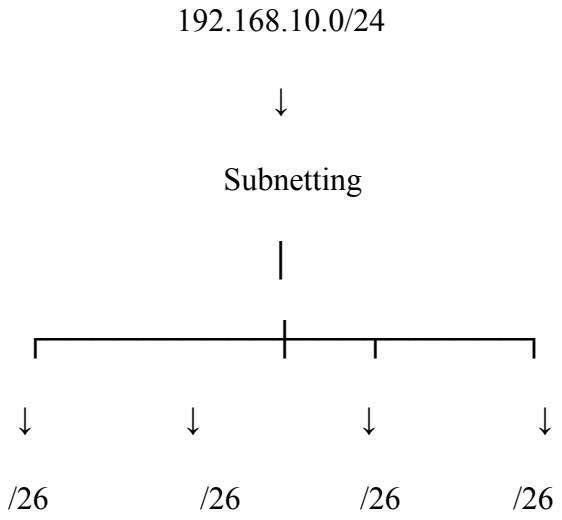
and wants separate networks for:

- Computer Lab
- Faculty
- Administration
- Management

Instead of putting all devices into one network, we can **divide the network into smaller networks**. This process is called **subnetting**.

> **Subnetting is the process of dividing one IP network into smaller logical networks called subnets.**

For example:



Understanding Network and Host Portions

An IPv4 address contains **32 bits**. For IP address 192.168.10.25/24. The /24 means:

- 24 bits → Network
- 8 bits → Host

The subnet mask is 255.255.255.0. In binary:

11111111.11111111.11111111.00000000

The 1s represent the **network portion**. The 0s represent the **host portion**.

What Does Subnetting Actually Do?

Suppose we start with:

192.168.10.0/24

Originally:

- Network bits = 24
- Host bits = 8

If we borrow 2 host bits for subnetting:

- Network bits = 26
- Host bits = 6

So the new prefix becomes /26. The subnet mask becomes:

255.255.255.192

We now have:

- $2^2 = 4$ subnets

Each subnet has:

- $2^6 = 64$ total addresses

and:

64 - 2 = 62 usable hosts

So:

Borrowing bits creates more subnets but leaves fewer hosts in each subnet.

This is one of the most important ideas in subnetting.

Subnetting Calculation Rules

Number of subnets = 2 ^ borrowed bits

For example, if 3 bits are borrowed:

Number of subnets = 2^3 = 8 subnets

Usable hosts = 2 ^ host bits - 2

For example, for /27:

Host bits = 32 - 27 = 5

Therefore:

2^5 - 2

= 32 - 2

= 30 usable hosts

The -2 accounts for:

- Network address
- Broadcast address

Common Subnet Masks

CIDR	Subnet Mask	Total Addresses	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

This table will be useful throughout both labs.

Network Address, Host Range and Broadcast Address

Consider:

192.168.10.25/24

The subnet is:

192.168.10.0/24

The addresses are:

192.168.10.0 → Network address

192.168.10.1 → First usable host

...

192.168.10.254 → Last usable host

192.168.10.255 → Broadcast address

Therefore:

Type	Address
Network	192.168.10.0
First Host	192.168.10.1
Last Host	192.168.10.254
Broadcast	192.168.10.255

Remember

- The **network address** identifies the subnet.
- The **broadcast address** is used to communicate with all hosts in that subnet.

The addresses between them are normally available for hosts.

Example: Finding a Subnet

Consider:

- Network: 192.168.10.0/24

We need 4 subnets.

1. Find the required borrowed bits. We need 4 subnets.

   $$2^2 = 4$$

   Therefore, borrow 2 bits.

2. Find the new prefix

   $$24 + 2 = 26$$

3. Therefore:

   $$/26$$

4. Find the subnet mask

   $$/26 = 255.255.255.192$$

5. Find the block size. Look at the last octet:

   $$256 - 192 = 64$$

   Therefore, the subnets increase by 64.

6. List the subnets
   - 0
   - 64
   - 128
   - 192

Therefore:

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.10.0	.1	.62	.63
2	192.168.10.64	.65	.126	.127
3	192.168.10.128	.129	.190	.191
4	192.168.10.192	.193	.254	.255

Each subnet has 62 usable hosts.

Example: Two subnets

Subnet mask

/25 = 255.255.255.128

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.126	192.168.10.127
2	192.168.10.128	192.168.10.129	192.168.10.254	192.168.10.255

Example: 8 subnets

Subnet mask

/27 = 255.255.255.224

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.30	192.168.10.31

2	192.168.10.32	192.168.10.33	192.168.10.62	192.168.10.63
3	192.168.10.64	192.168.10.65	192.168.10.94	192.168.10.95
4	192.168.10.96	192.168.10.97	192.168.10.126	192.168.10.127
5	192.168.10.128	192.168.10.129	192.168.10.158	192.168.10.159
6	192.168.10.160	192.168.10.161	192.168.10.190	192.168.10.191
7	192.168.10.192	192.168.10.193	192.168.10.222	192.168.10.223
8	192.168.10.224	192.168.10.225	192.168.10.254	192.168.10.255

How to Find Which Subnet an IP Belongs To

Suppose we have:

- IP: 192.168.10.75
- Mask: 255.255.255.192

Our /26 subnets are:

- 0 – 63
- 64 – 127
- 128 – 191
- 192 – 255

The value 75 falls between 64 – 127. Therefore:

- Network: 192.168.10.64
- First Host: 192.168.10.65
- Last Host: 192.168.10.126
- Broadcast: 192.168.10.127

So:

192.168.10.75 belongs to the 192.168.10.64/26 subnet.

Activity 01 — Calculate Subnet Information

Divide the network into **4 equal subnets**. Determine:

- New prefix
- Subnet mask
- Number of subnets
- Usable hosts per subnet
- Network address
- First host
- Last host
- Broadcast address

Given:

192.168.20.0/24

We need **4 equal subnets**.

- **Step 1: Find the new prefix**
  - To create 4 subnets:
    - $2^2 = 4$
    - Borrow **2 bits** from the host portion.
  - Original: /24
  - New: /26

- **Step 2: Subnet mask**
  - /26 = 255.255.255.192

- **Step 3: Hosts per subnet**
  - There are:
    - 32 - 26 = 6 host bits
  - Total addresses: $2^6 = 64$
  - Usable hosts: 64 - 2 = 62

- So each subnet has **62 usable hosts**.

- **Step 4: Find the subnets**
  - The block size is 256 - 192 = 64
  - Therefore, the networks start at: 0, 64, 128, 192

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.20.0	192.168.20.1	192.168.20.62	192.168.20.63
2	192.168.20.64	192.168.20.65	192.168.20.126	192.168.20.127
3	192.168.20.128	192.168.20.129	192.168.20.190	192.168.20.191
4	192.168.20.192	192.168.20.193	192.168.20.254	192.168.20.255

Final answers:

- **New prefix:** /26
- **Subnet mask:** 255.255.255.192
- **Number of subnets:** 4
- **Usable hosts per subnet:** 62

Activity 02 — Identify the Subnet

For each IP address, determine its subnet information.

IP Address	Mask	Network	First Host	Last Host	Broadcast
192.168.10.25	/26	_____	_____	_____	_____
192.168.10.75	/26	_____	_____	_____	_____

192.168.10.140	/26	_____	_____	_____	_____
192.168.10.200	/26	_____	_____	_____	_____

Solution:

Given /26. The subnet mask:

255.255.255.192

The block size is:

256 - 192 = 64

Therefore, the address ranges are:

- 192.168.10.0 – 192.168.10.63
- 192.168.10.64 – 192.168.10.127
- 192.168.10.128 – 192.168.10.191
- 192.168.10.192 – 192.168.10.255

Solution

IP Address	Mask	Network	First Host	Last Host	Broadcast
192.168.10.25	/26	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
192.168.10.75	/26	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
192.168.10.140	/26	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
192.168.10.200	/26	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

Quick way to identify the subnet

For /26, look at the **last octet**:

- 0–63 → Network 0
- 64–127 → Network 64
- 128–191 → Network 128
- 192–255 → Network 192

For example:

192.168.10.140

↑

140

140 falls between **128 and 191**, so:

- Network = 192.168.10.128
- Broadcast = 192.168.10.191

From Subnet Design to Packet Tracer

We have learned how to calculate subnet ranges. Now we will use those calculations to build actual networks.

The process is:

Subnet Design → Create Topology → Assign IP Addresses → Configure PCs

↓

Configure Router

↓

Verify Interfaces

↓

Test Connectivity

The important idea is:

> Packet Tracer does not calculate your network design for you. You must determine the correct subnet addresses first and then configure the devices accordingly.

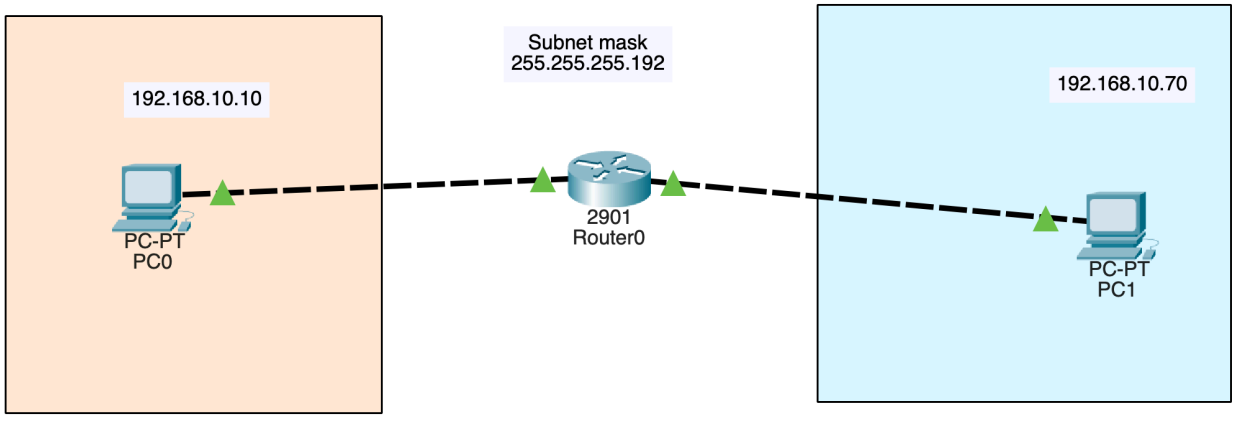
Implementation

Task 01: Ping between two subnets

Use:

- Subnet A → 192.168.10.0/26
- Subnet B → 192.168.10.64/26

Topology



Addressing Plan

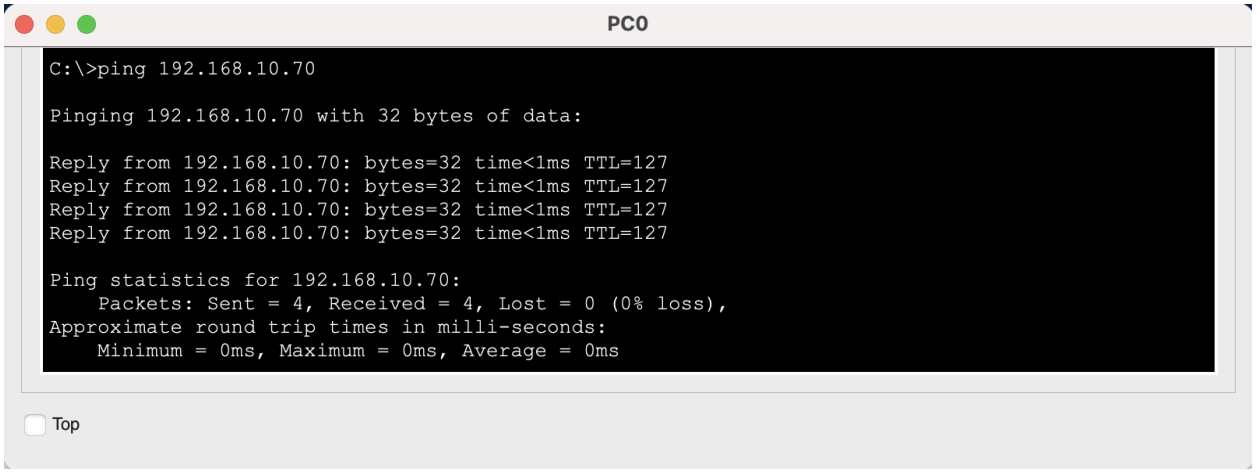
Device	IP Address	Mask	Gateway
PC0	192.168.10.10	255.255.255.192	192.168.10.1

Router G0/0	192.168.10.1	255.255.255.192	—
Router G0/1	192.168.10.65	255.255.255.192	—
PC1	192.168.10.70	255.255.255.192	192.168.10.65

Tasks

1. Place two PCs and one router.
2. Connect the devices.
3. Configure PC0.
4. Configure PC1.
5. Configure G0/0.
6. Configure G0/1.
7. Activate both interfaces using no shutdown.
8. Verify the interfaces.
9. Test PC0 → PC1.

Ping from PC0 to PC1



For broadcasting, you can use the last ip address of each subnet.

Task 02: Create a small network and try broadcasting

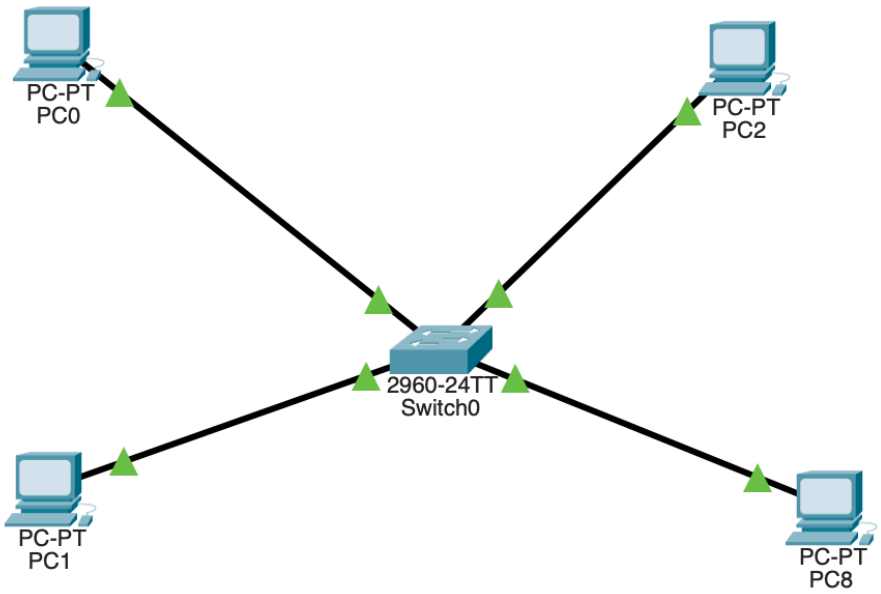
192.168.10.10

IP Addressing

Device	IP Address	Subnet Mask
PC-0	192.168.10.10	255.255.255.0
PC-1	192.168.10.40	255.255.255.0
PC-2	192.168.10.120	255.255.255.0
PC-3	192.168.10.140	255.255.255.0

Tasks

1. Add **4 PCs** and **1 switch**.
2. Connect all PCs to the switch.
3. Configure the IP addresses.
4. Configure the subnet mask.
5. From PC-0, ping all PCs.
6. Record the results.



Task 03: Calculate Subnet Information and then implement it in Cisco Packet Tracer

Given:

192.168.30.0/24

Divide the network into **2 equal subnets**.

Determine:

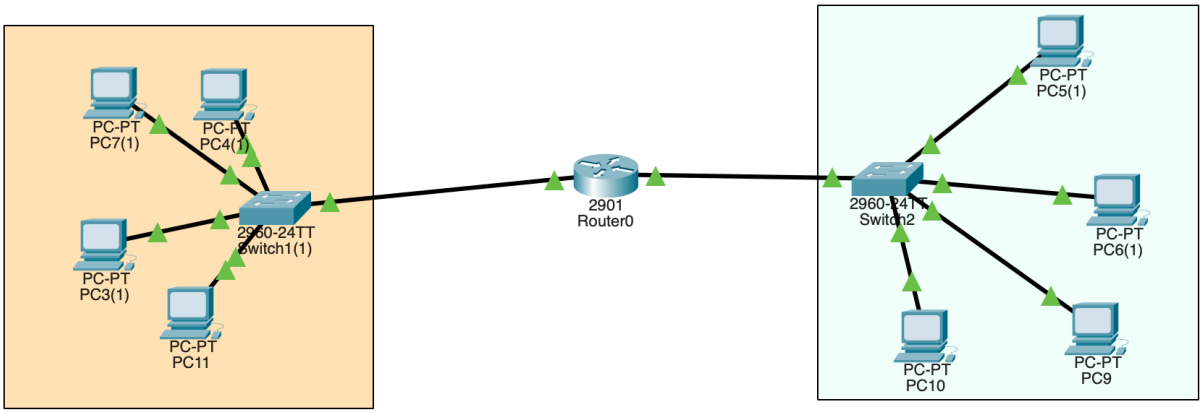
- New prefix
- Subnet mask
- Number of subnets
- Usable hosts per subnet
- Network address
- First host
- Last host
- Broadcast address

Complete:

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.126	192.168.10.127
2	192.168.10.128	192.168.10.129	192.168.10.254	192.168.10.255

Tasks

1. Add **4 PCs** and **1 switch in each subnet**.
2. Connect both switches to one router.
3. Configure the IP address and subnets.
4. Activate both interfaces using no shutdown.
5. Verify the interfaces.
6. Ping between PCs in the same network.
7. Broadcast in Network A and then in Network B.
8. Ping from Network A and B.
9. Record the results.


PC-PT
PC7(1)
PC-PT
PC4(1)
2960-24TT
Switch1(1)
PC-PT
PC3(1)
PC-PT
PC11
2901
Router0
2960-24TT
Switch2
PC-PT
PC5(1)
PC-PT
PC6(1)
PC-PT
PC10
PC-PT
PC9